

Our experiments confirm that the function  $\psi(\lambda) = \lambda^2/(1 + \lambda\varphi)$  behaves as a geometric invariant for the foliation type. However, Eq. (7) reveals an unexpected resonance near  $\lambda = \varphi^2 \approx 2.618$ . ~~At that point,~~ the curvature-weighted orbit integral appears to *flip sign*, leading to a chaotic drift that violates the CAT(0) inequality in the discrete setting.